Tobias Mwangi-Hartley

PROFESSIONAL SUMMARY

Data Scientist transitioning into Machine Learning Engineering, with experience building, evaluating, and operating scheduled forecasting and batch-scoring workflows in Python and SQL on GCP. Delivered measurable improvements through a recipe-demand forecasting workflow and churn-targeting analysis. Experienced in feature engineering, orchestration through scheduled jobs, Docker, pytest, and automated tests for forecasting feature code. MSc in Statistics, with a strong foundation in modelling and experimental design.

TECHNICAL SKILLS

Programming and analysis: Python, SQL, R
Machine learning: LightGBM, XGBoost, scikit-learn, feature engineering, SHAP, forecasting, churn modelling
Cloud: GCP — BigQuery, Vertex AI Workbench; dbt, dbt Cloud Scheduler, Looker
Orchestration: scheduled notebook jobs, dbt Cloud Scheduler
Engineering and experimentation: pytest, unit testing, Docker, Streamlit, statsmodels, R Markdown, sequential A/B testing
Additional Training and Projects: PyTorch (course project), torchvision, Google Colab

PROFESSIONAL EXPERIENCE

Data Scientist | Thistlebox *2022–Present*

- Built a six-week-ahead weekly recipe-demand forecasting workflow using LightGBM, with calendar, price and menu-position features.
- Orchestrated scheduled batch forecasting through weekly notebook jobs in Vertex AI Workbench; wrote forecast outputs to BigQuery for operational reporting in Looker.
- Improved forecast WAPE from 23% to 14%; estimated food waste reduction was 9%.
- Developed and ran a weekly XGBoost churn-scoring workflow in BigQuery to batch-score customers and identify the top decile for retention offers.
- Validated retention targeting using a 10% holdout; churn decreased by 2.1 percentage points in the treated group.
- Built and containerised a Python Streamlit application; wrote the Dockerfile, while the data platform team deployed it to Cloud Run. Used by approximately 25 people in product and growth.
- Added pytest unit tests for forecasting feature-engineering code in 2025, reaching approximately 60% test coverage.
- Delivered the internal training session “Stats for PMs” three times.

Junior Data Analyst | Firth Energy Retail *2020–2022*

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

EDUCATION

MSc in Statistics
Dissertation: Bayesian hierarchical models

ADDITIONAL PROJECT

Image Classification Course Project | Personal *2025*

- Fine-tuned a small CNN in PyTorch on a public food-images dataset as a learning project, as part of an online deep-learning course, achieving approximately 80% accuracy.
- Tools: PyTorch, torchvision, Google Colab.